TM - DATA COLLAR

Development of an V2 intelligent data collar for primate monitoring using the nRF54L15 and Zephyr RTOS.

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CONTEXT

Scientific need

- Observe Vervet behavior
- Acoustic records
- 24/7 monitoring

The solution : nRF Monkey

- Continuous recording to SD card
- Management via mobile app (BLE)
- Remote release

Field challenge

- Weather and terrain
- Limited size & weight
- Battery life

V1 LIMITATIONS

- Power consumption
- Data overload
- Social context
- Mechanical

OBJECTIVES

1. Hardware migration
2. Intelligent sound level detection
3. Proximity monitoring
4. Profiling & validation
5. Consumption optimization

DESIGN

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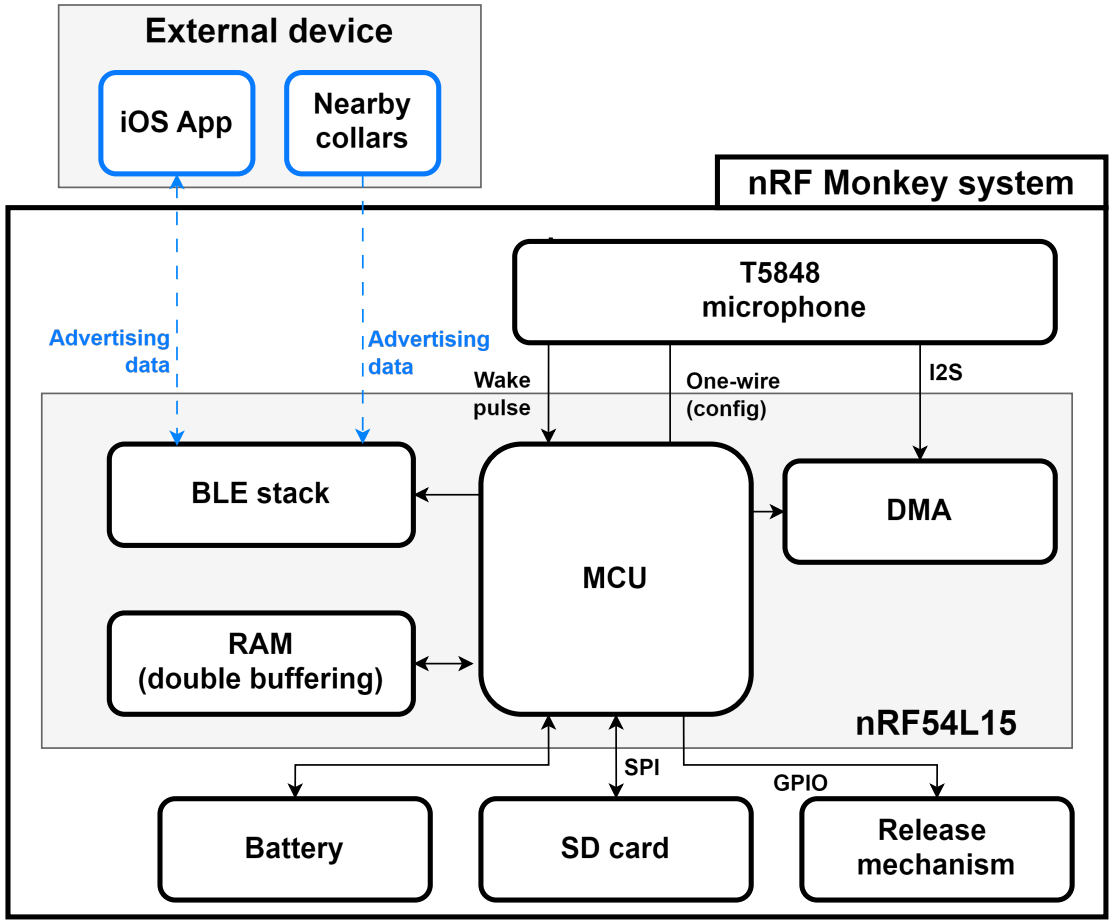
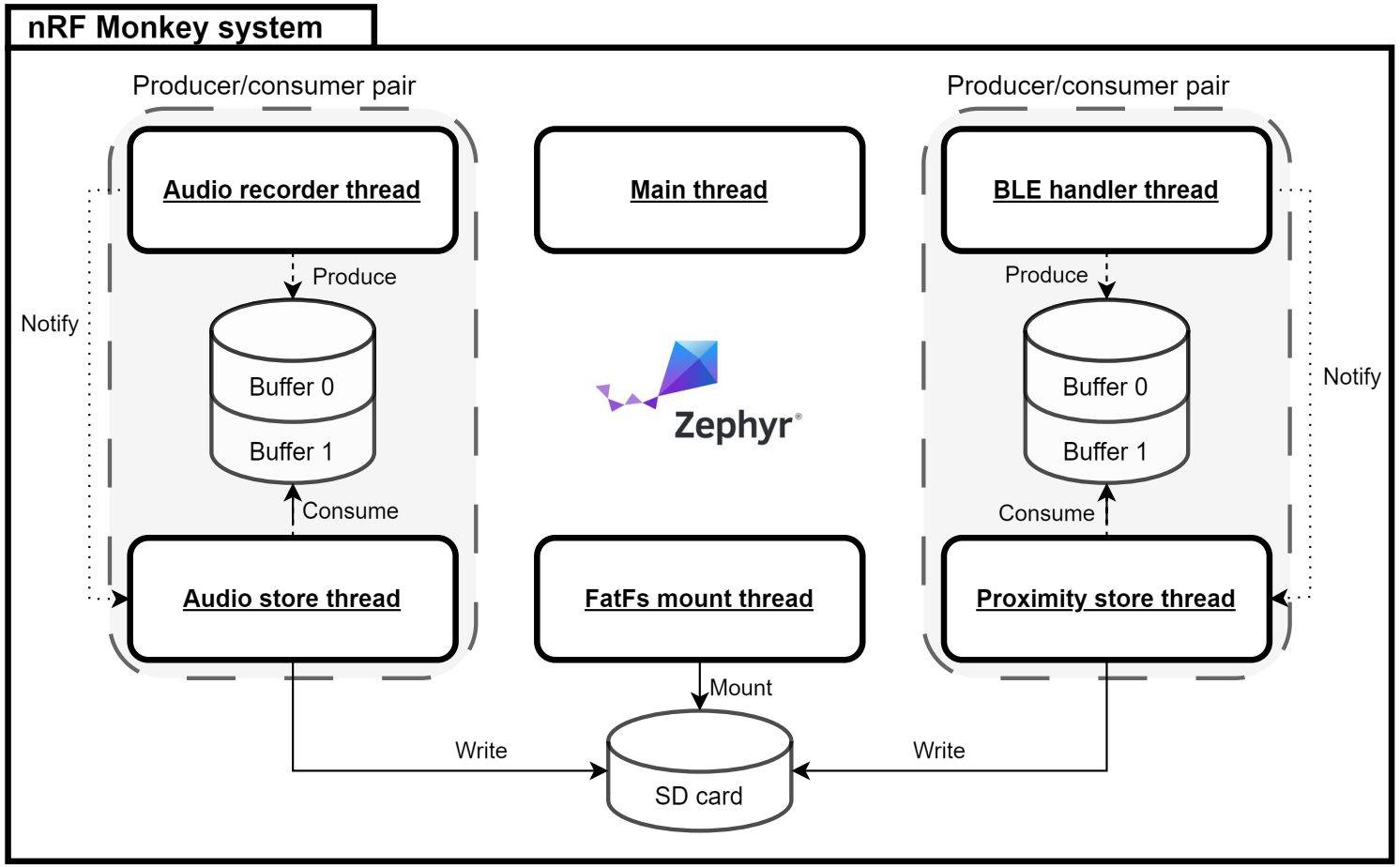
OVERVIEW

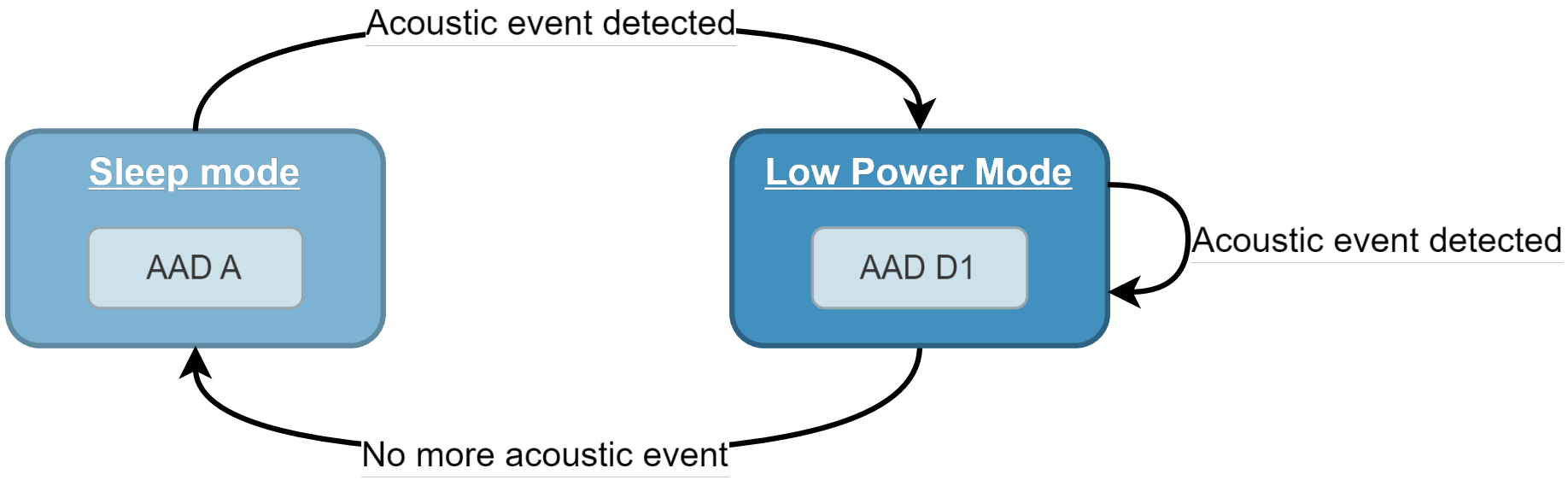


FIRMWARE


nRF Monkey system
Producer/consumer pair
Audio recorder thread
Produce
Buffer 0
Buffer 1
Consume
Audio store thread
Notify
Main thread
Zephyr
FatFs mount thread
Mount
SD card
Write
Write
Producer/consumer pair
BLE handler thread
Produce
Buffer 0
Buffer 1
Consume
Proximity store thread
Notify

INTELLIGENT RECORDING

MICROPHONE



* AAD = Acoustic Activity Detection

Consumption : 20 μA in *Sleep Mode* | 120 μA in *Low Power Mode* (V1 microphone: 600 μA)

AAD ALGORITHM

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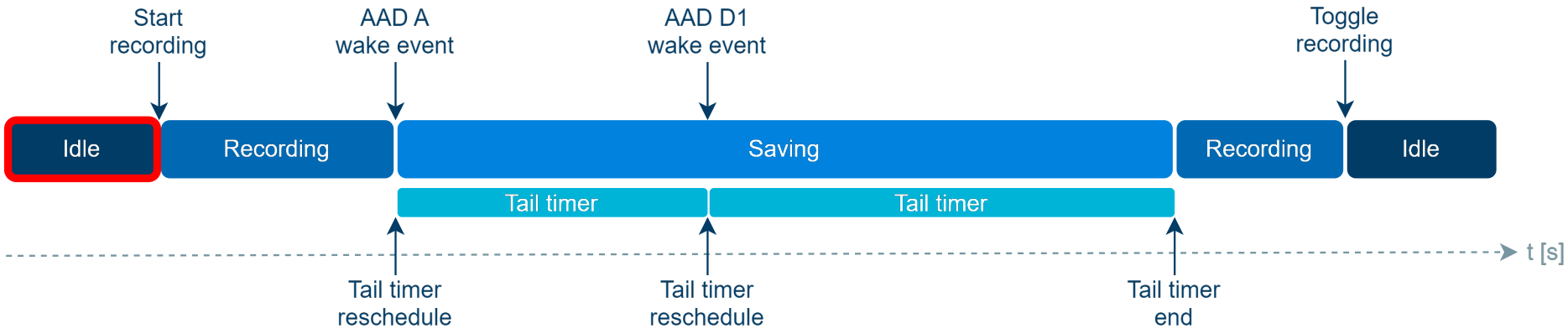
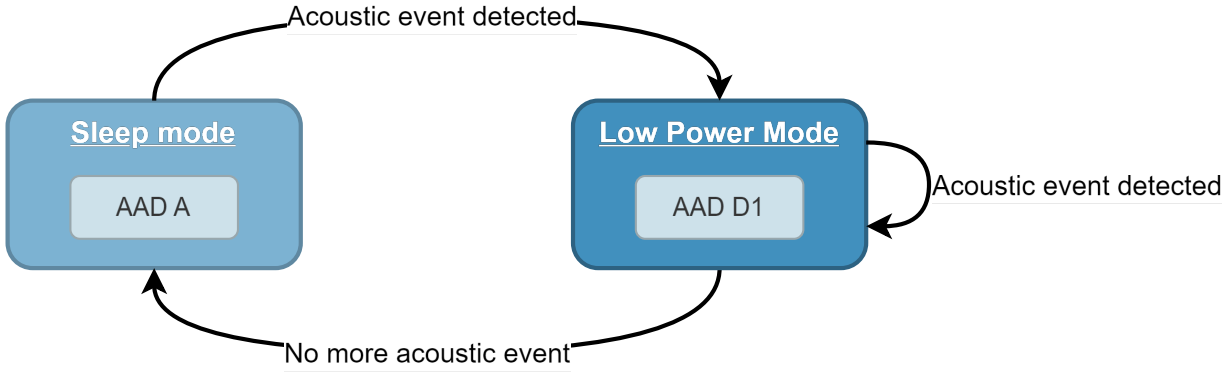
RECORDING PROCESS





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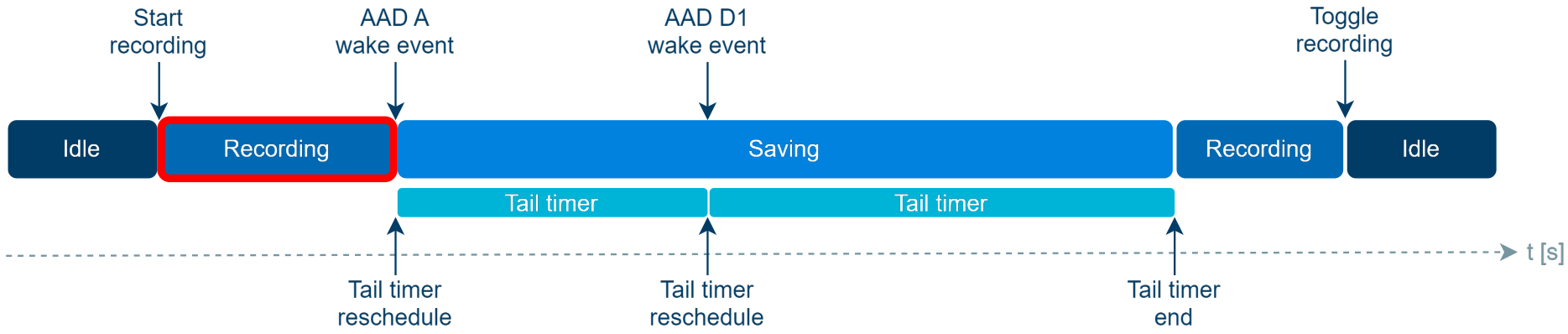

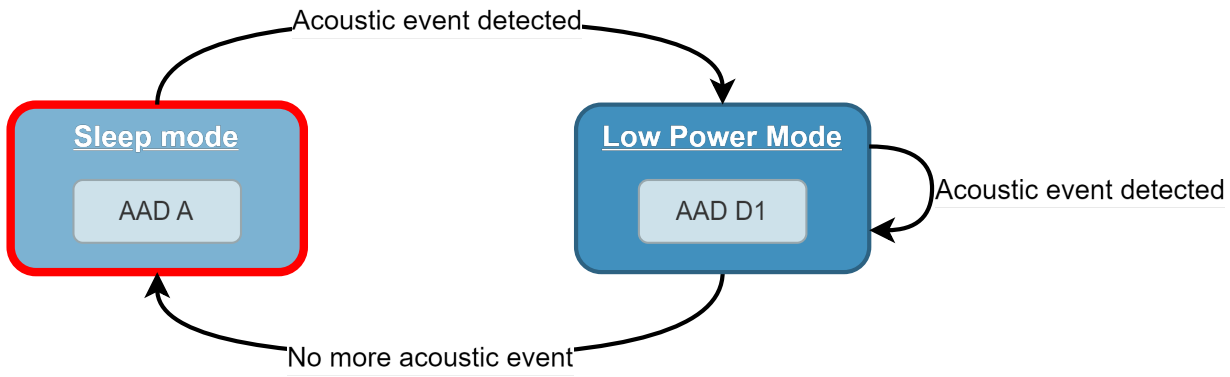

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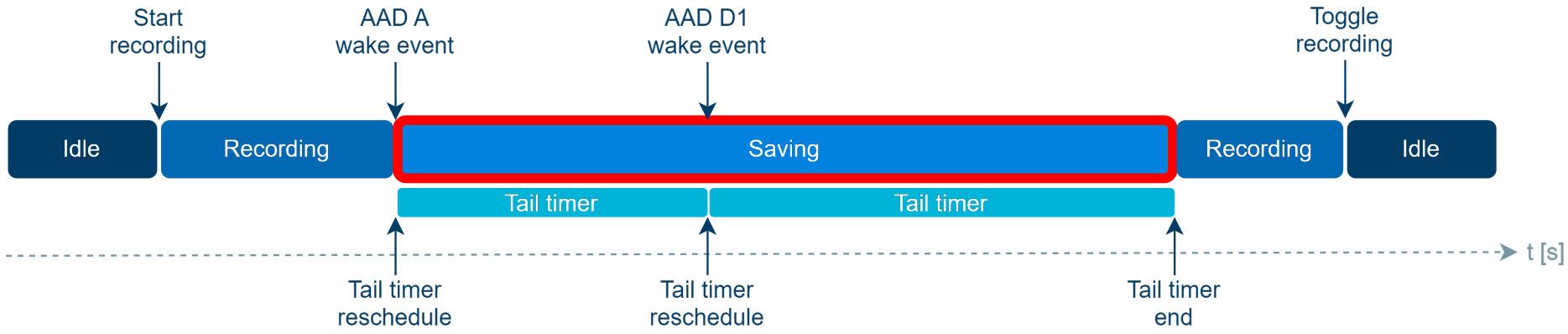
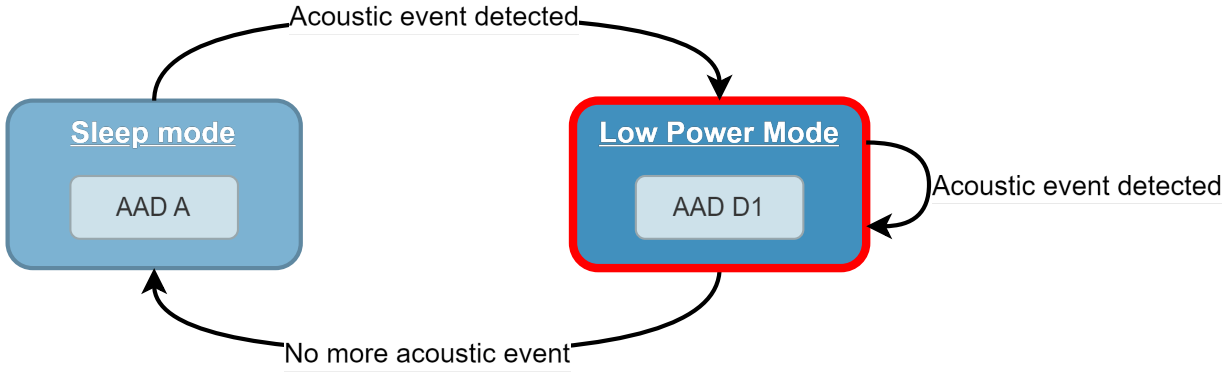
RECORDING PROCESS





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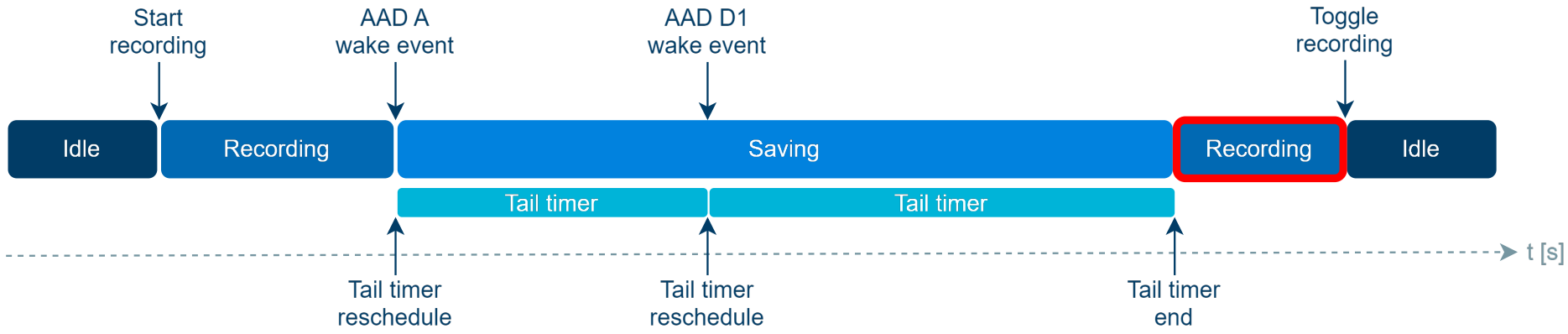

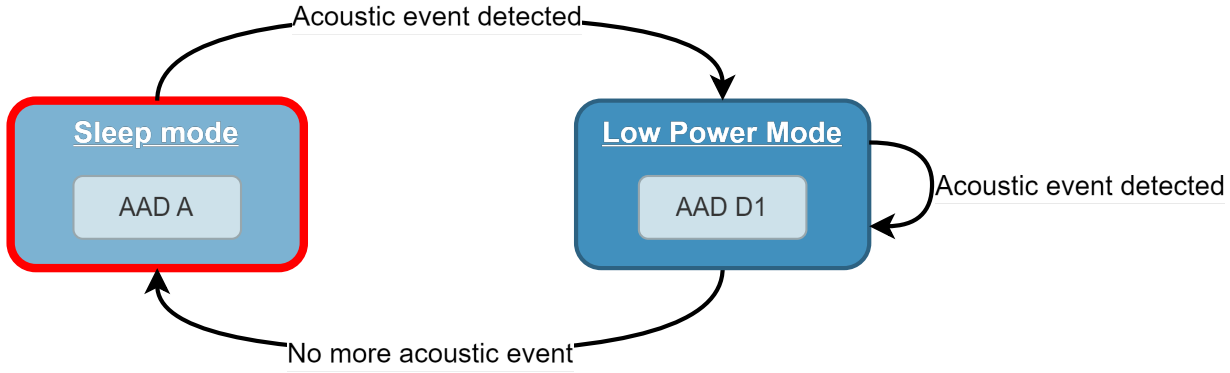

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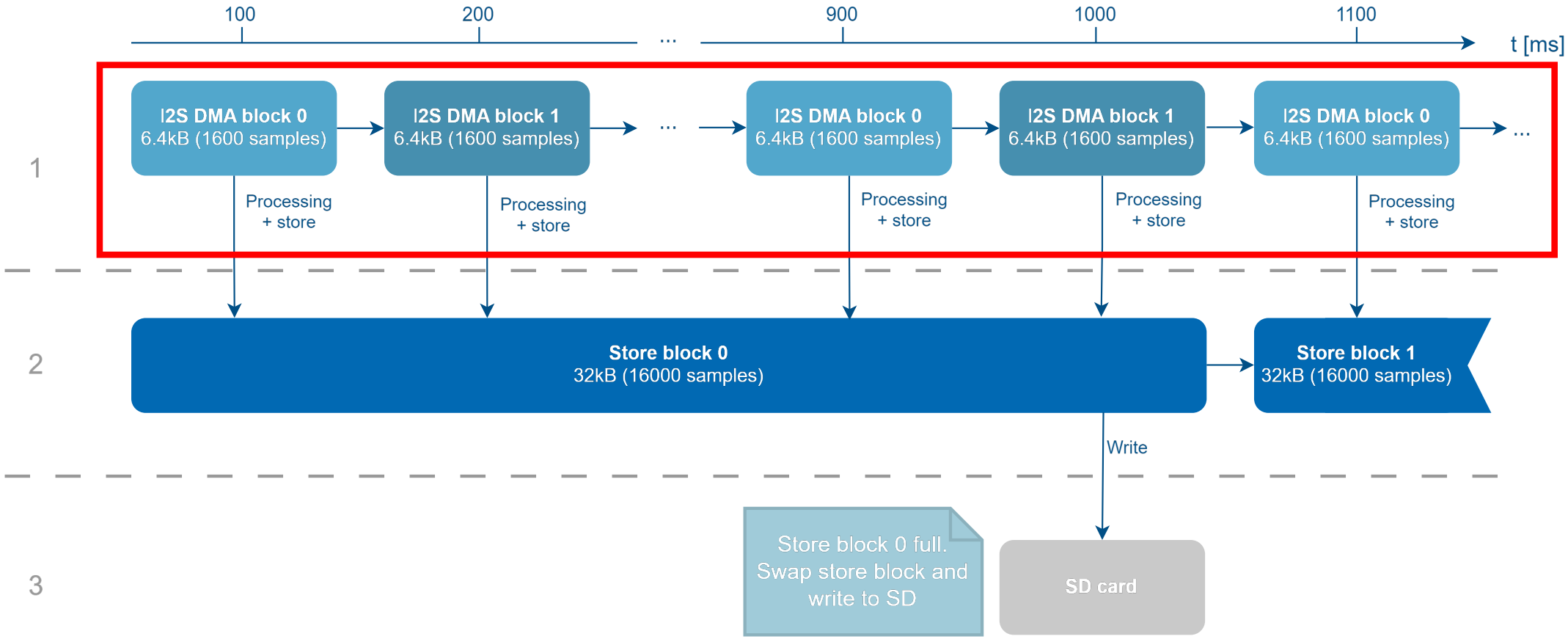
RECORDING MEMORY MANAGEMENT



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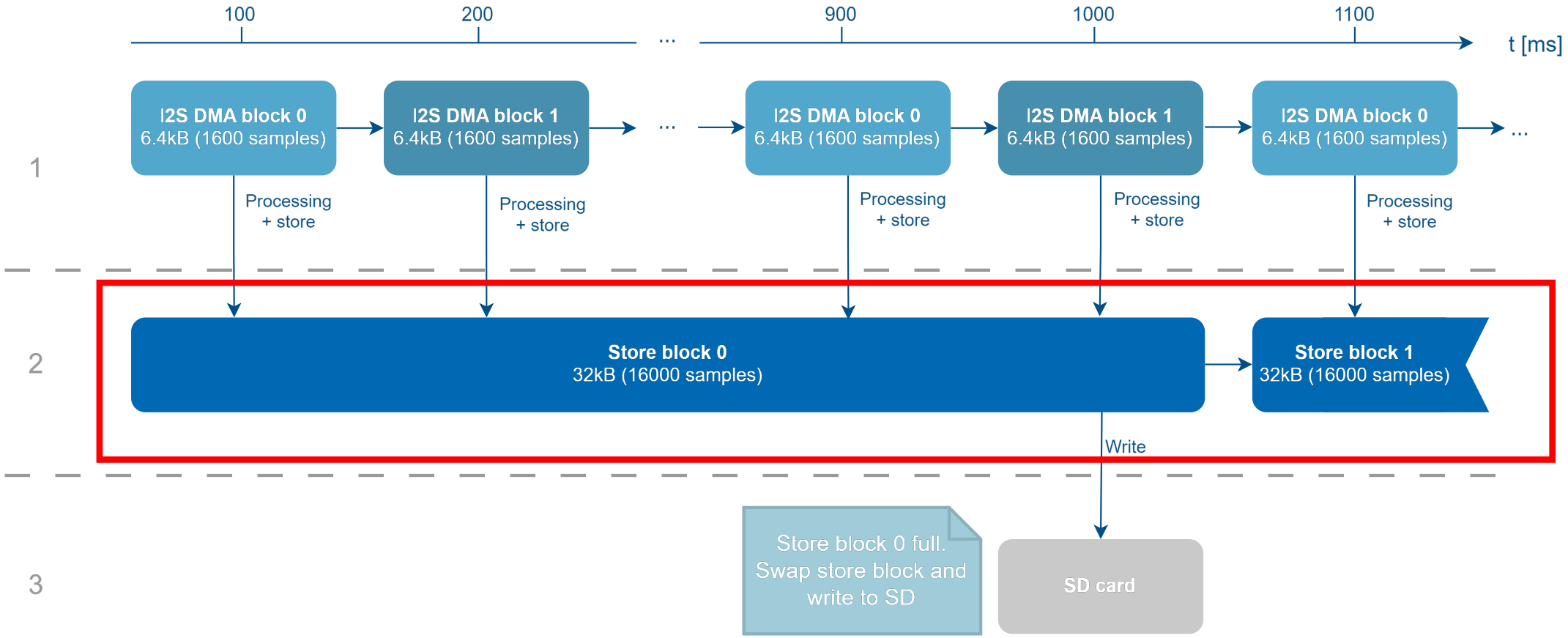
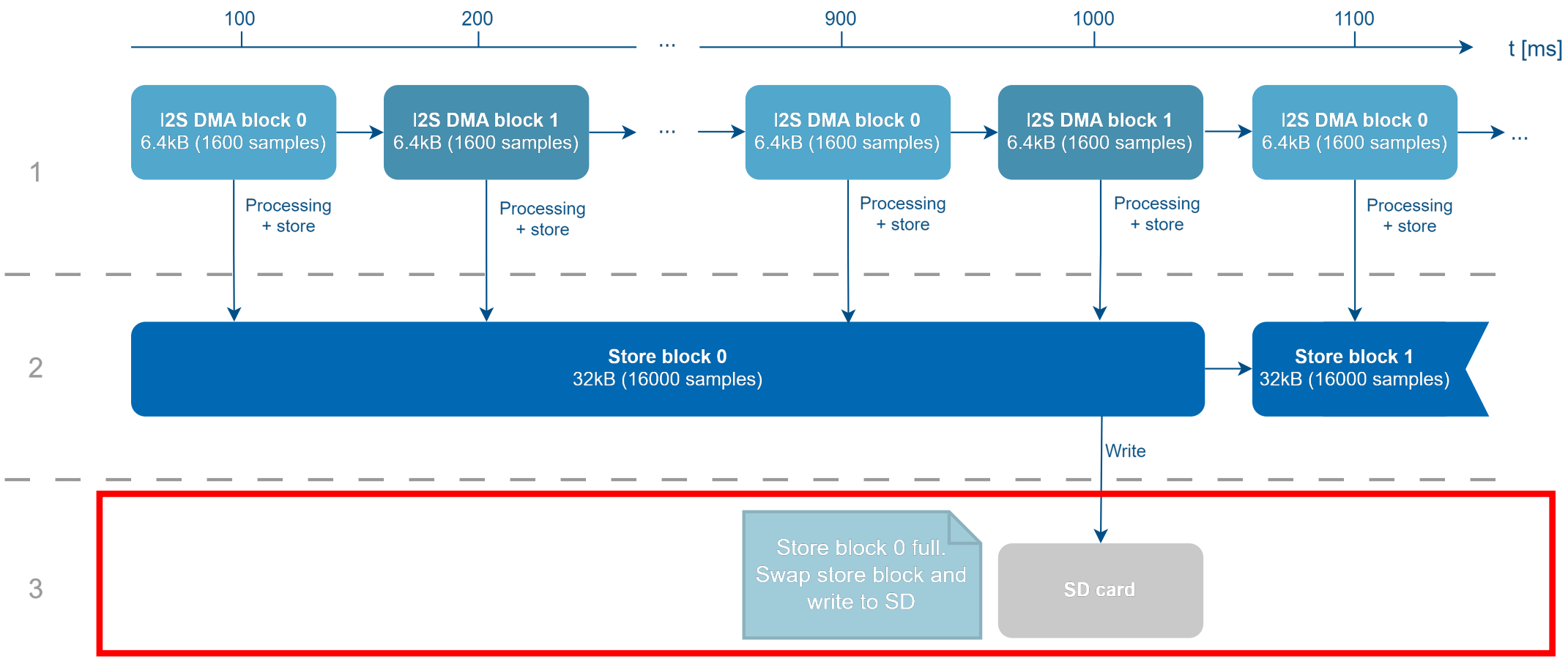
RECORDING MEMORY MANAGEMENT



RECORDING MEMORY MANAGEMENT



PROXIMITY DETECTION

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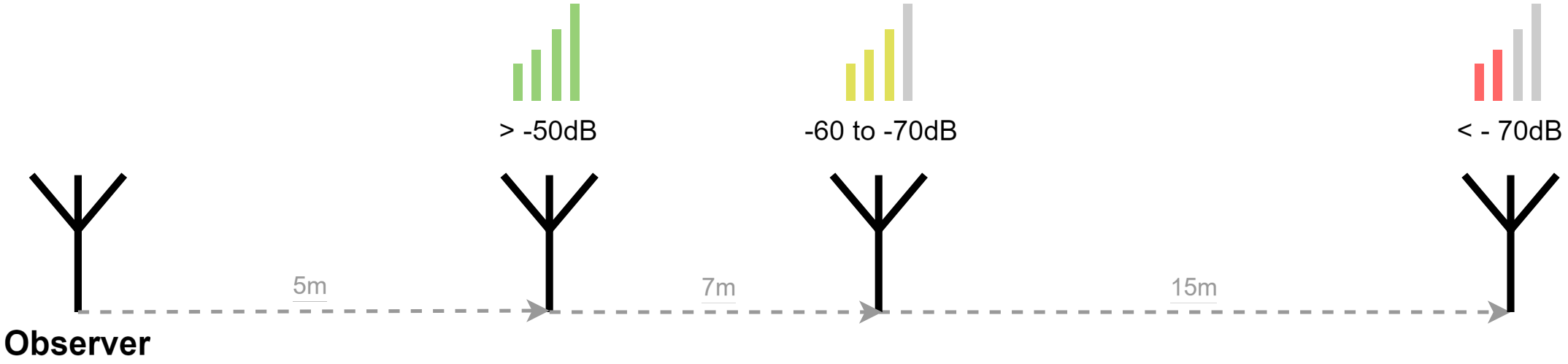
BLE RSSI (RECEIVED SIGNAL STRENGTH INDICATOR)

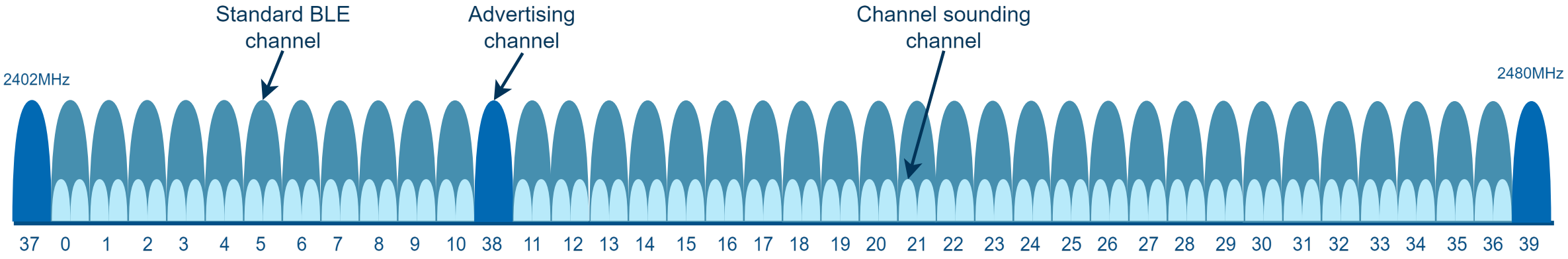
Estimates distance based on the power level of the received signal.



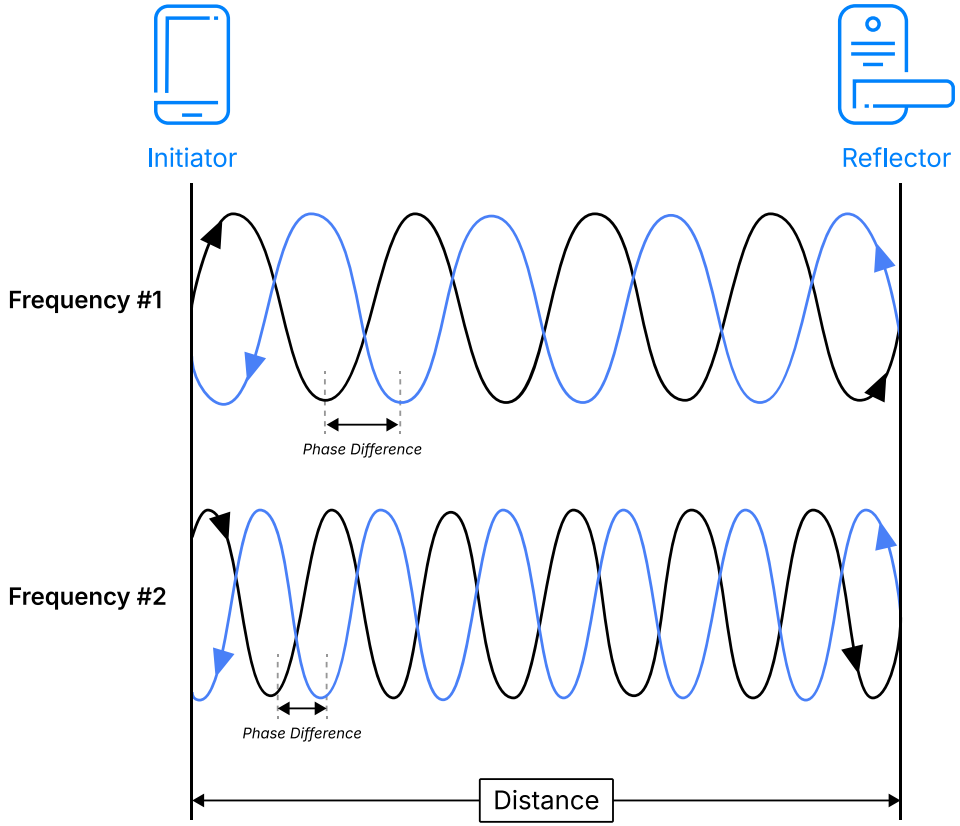
BLE CHANNEL SOUNDING I

- Introduce by Bluetooth® 6.0
- A high-precision distance measurement technique
- Use phased-based ranging (PBR) and round-trip time (RTT) to compute the distance

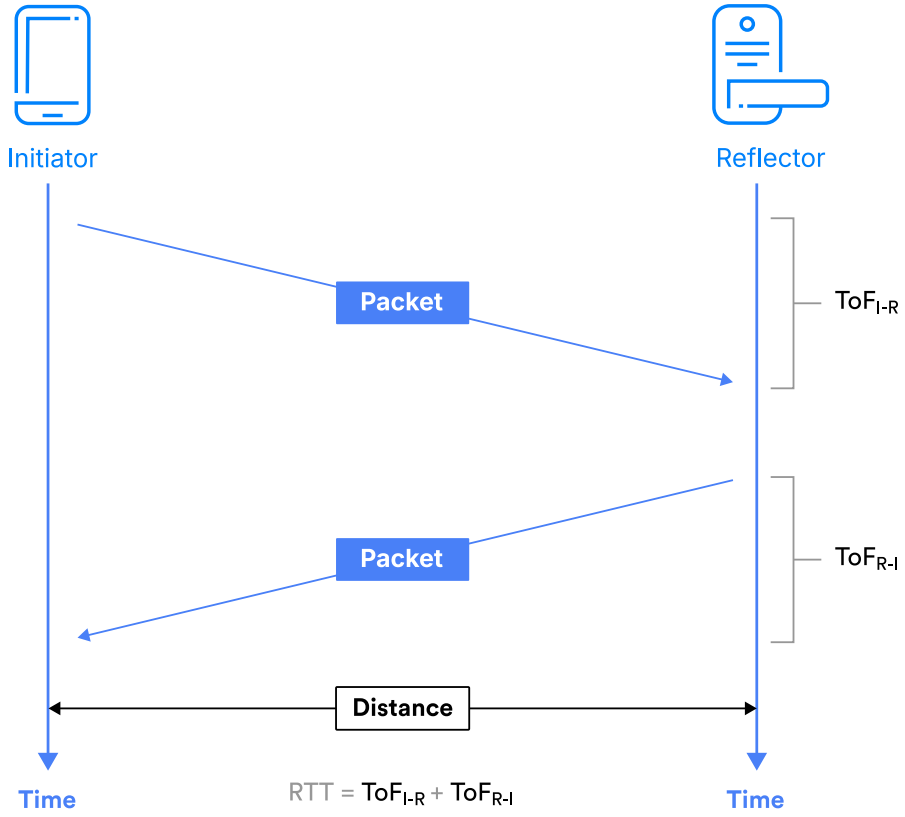
<u>CS channelization :</u>



BLE CHANNEL SOUNDING II



Phased-based ranging



Round-trip time

METHODS EVALUATION

Accuracy

Detection range

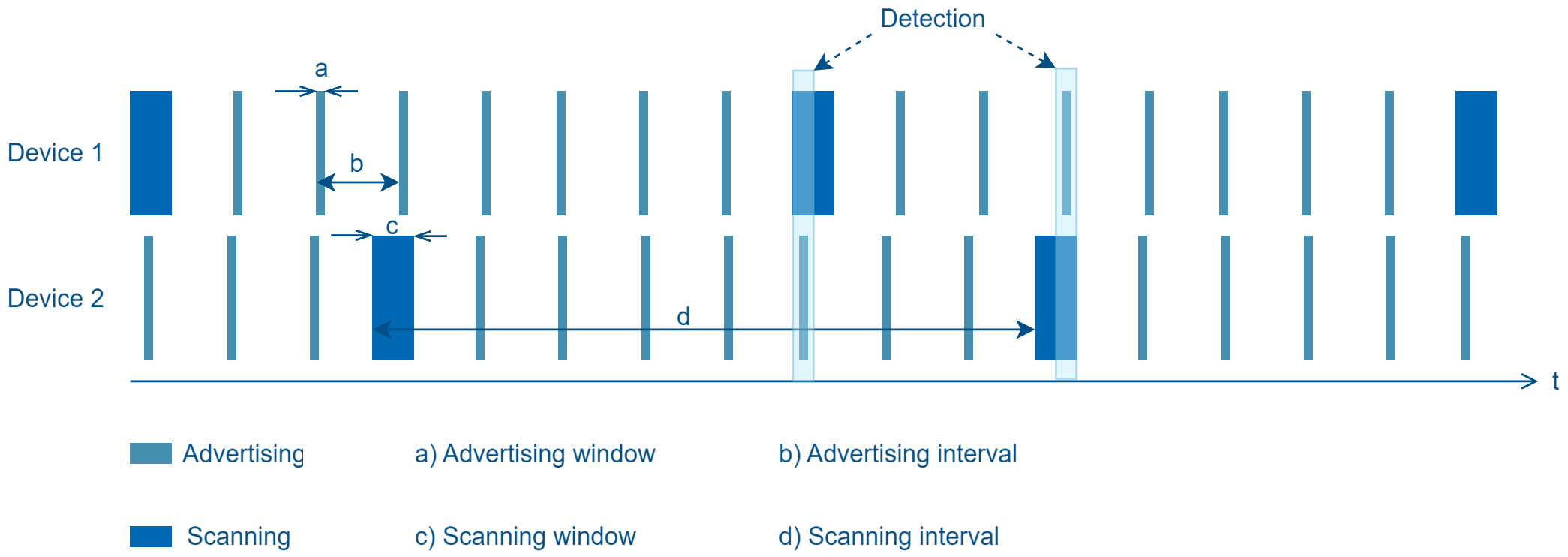
Power consumption

Nbr of measure

RESULTS SUMMARY

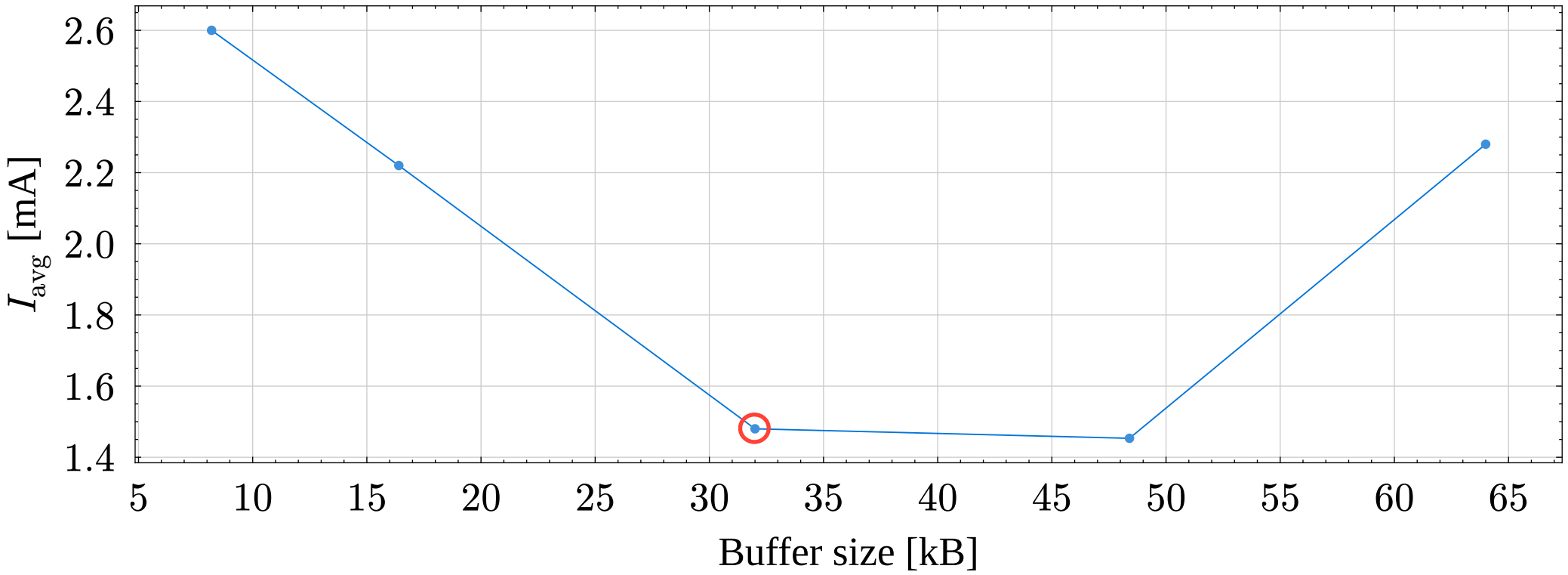
Metric	RSSI	Channel Sounding
Indoor overall mean error	3.29 m	0.98 m
Outdoor overall mean error	1.90 m	0.44 m
Max range	100 m (109 m error)	100 m (5.9 m error)
Power (active/isolated)	1.78 mA / 1.78 mA	0.63 mA / 1.78 mA
Max. measure	High (passive)	Limited (serial ACL)

RSSI PROXIMITY DETECTION PROCESS

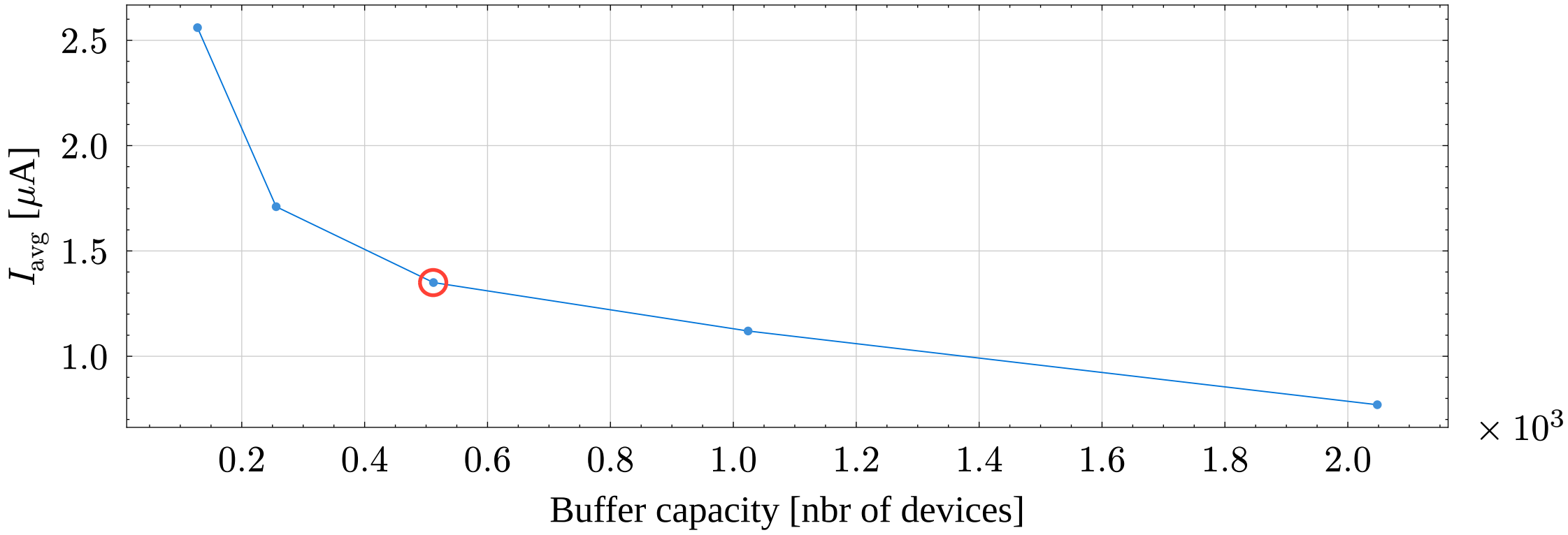


CONSUMPTION OPTIMIZATION

AUDIO BUFFER SIZE



PROXIMITY BUFFER SIZE

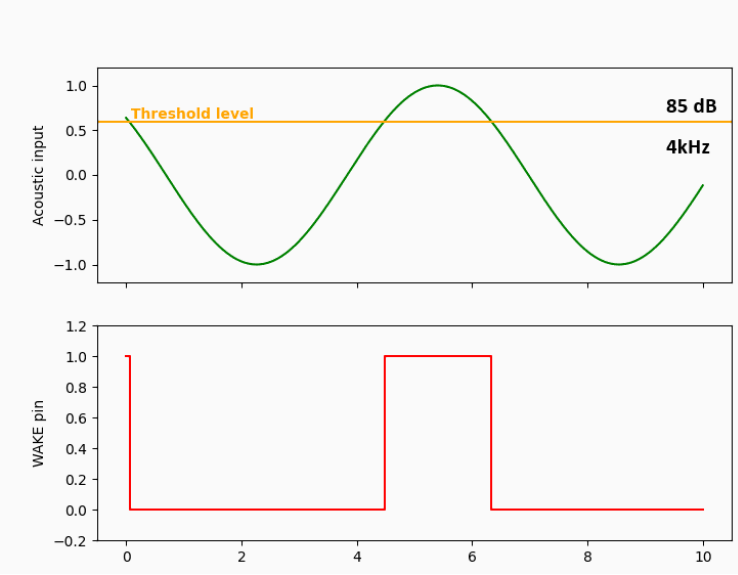


MICROPHONE PARAMETERS

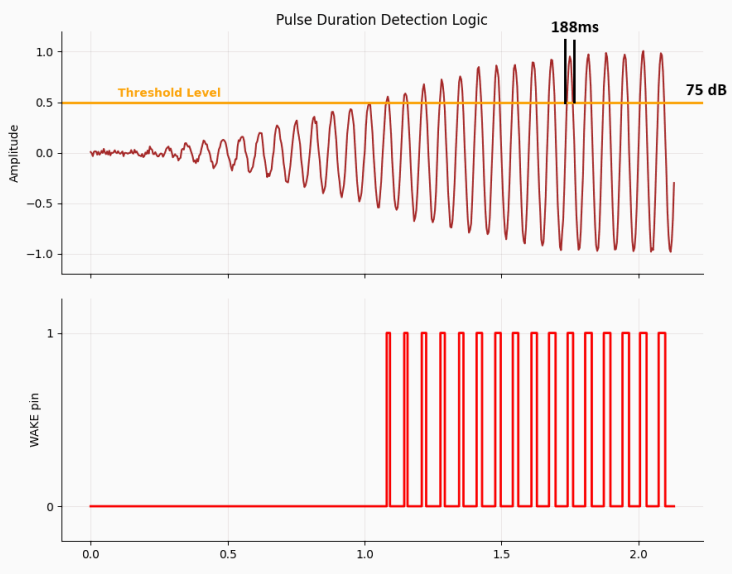
Primate vocalizations

- Frequency : 150–500 Hz & 2–3 kHz
- Duration : ~2s (multiple repetition with ~2s interval)
- Target level : 90–100 dB SPL
- Ambient : 65 dB SPL

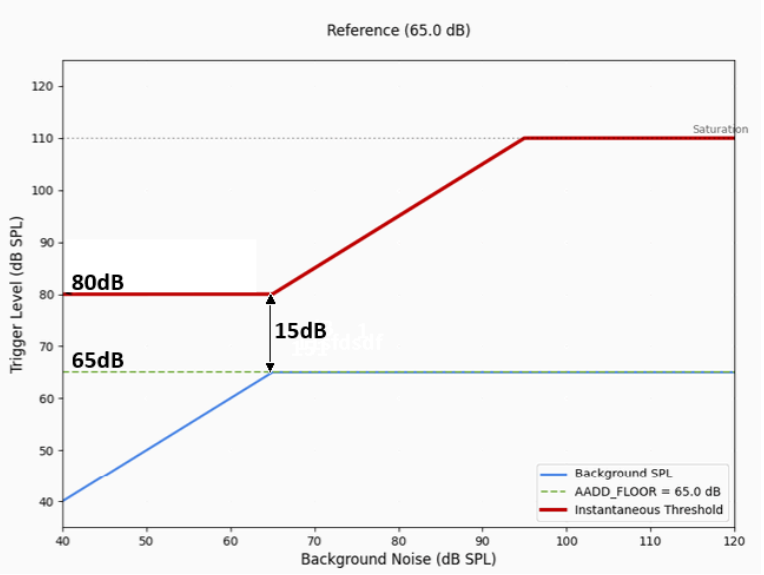
AAD analogique



AAD digital 1 - absolute

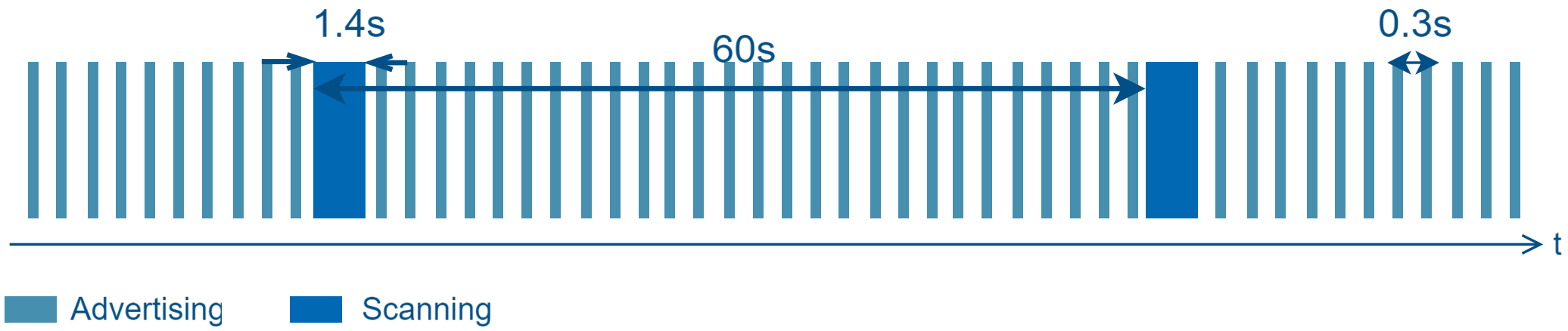


AAD digital 1 - relative



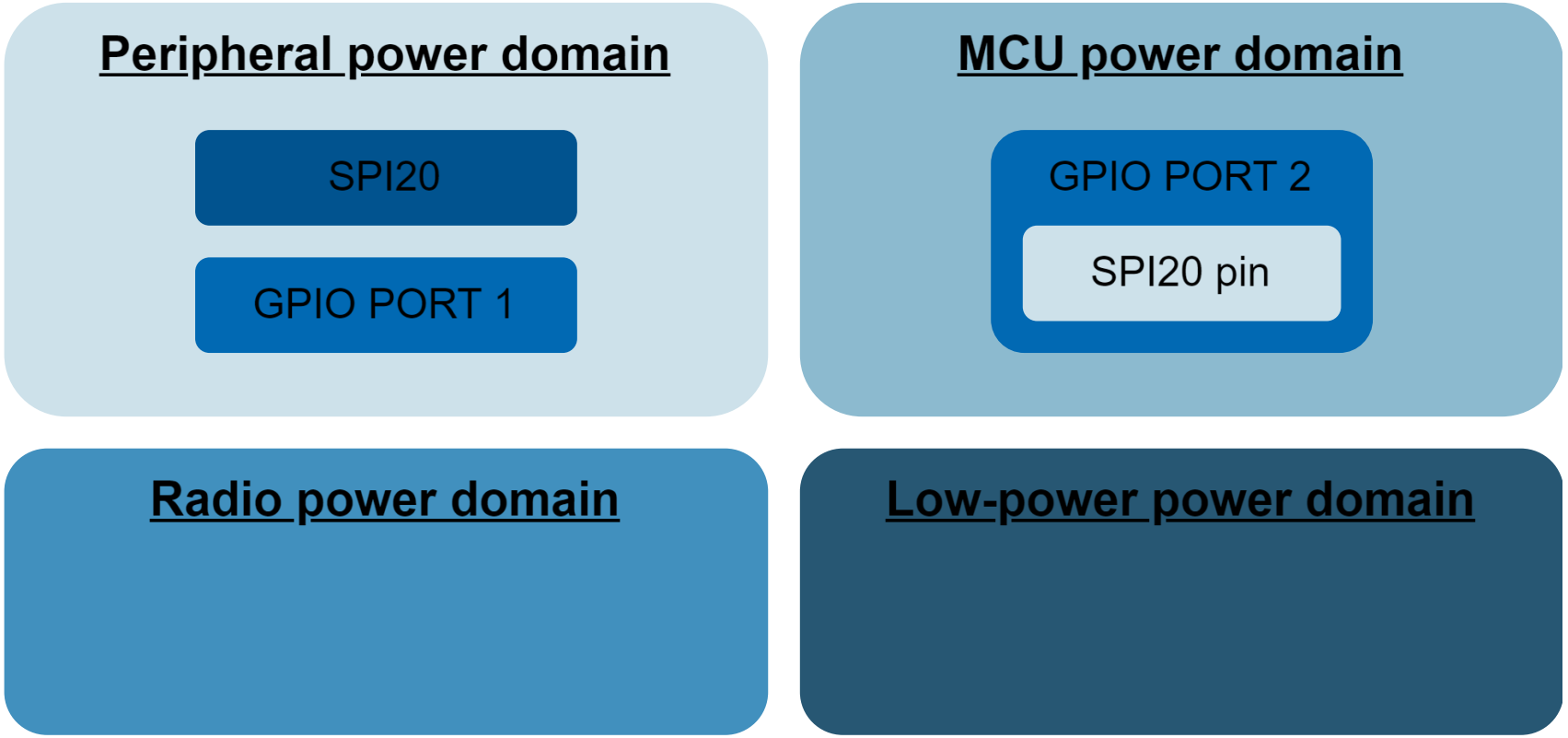
BLE PROXIMITY PARAMETERS

Theoretical modeling > Python simulation > Configuration test

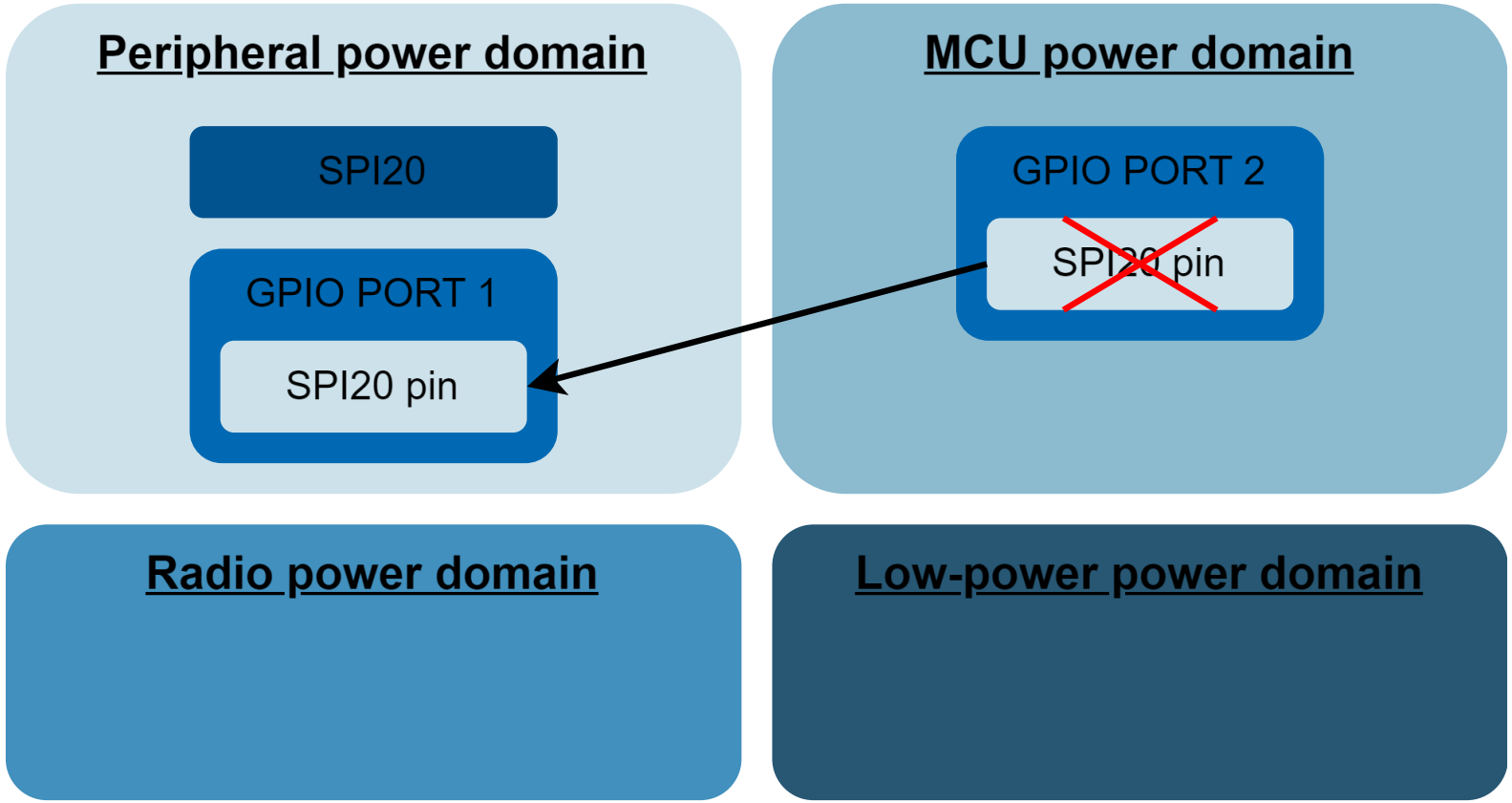


Mean current = 86 μA

POWER DOMAINS - MIC/SD BOARD I



POWER DOMAINS - MIC/SD BOARD II



CONSUMPTION SUMMARY

Mode	V1	V2
Saving	4.02 mA	**2.13 mA**
Recording	4.02 mA	**389.03 μA**

Batterie life estimation (2200mAh): **~ 220 days** with 1min/h recording

PROJECT CONCLUSION

1. Hardware migration ✓

2. Intelligent sound level detection ✓

3. Proximity monitoring ✓

4. Profiling & validation ✓

5. Consumption optimization ✓

FUTURE PERSPECTIVES

- Hardware
- Field tuning
- Mobile application
- Base station

SELF-ASSESSMENT

Zephyr RTOS

Low power design

Wireless protocols

Project management

QUESTIONS